

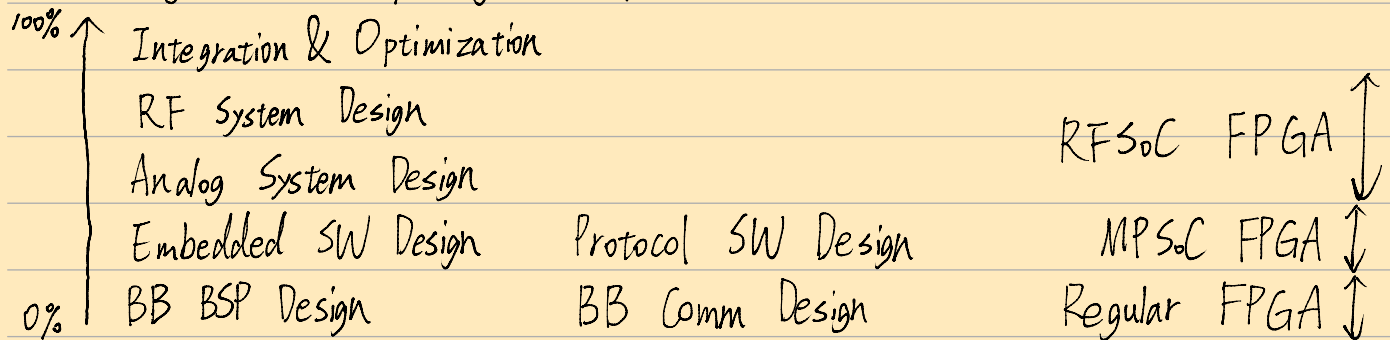
Ultra Lightweight Dehaze Methods for Robot Vision Using HLS

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* FPGA trends:

FPGA $\Rightarrow$ SoPC $\Rightarrow$ SoC FPGA $\Rightarrow$ Cloud FPGA $\Rightarrow$ RFSoc

* Heterogeneous Computing FPGA



* PYNQ Heterogeneous Hardware Acceleration

- Python productivity for ZYNQ
- An open-source project from Xilinx
- Designed by University of Texas at Austin

* High-Level Synthesis

C, C++, SystemC Directives



Vivado HLS



VHDL · Verilog · SystemC



RTL Export

IP-XACT · Sys Gen · PCore

HLS (Autopilot) Results: Sphere Decoder

HLS (Autopilot) Results: QRD

HLS (Autopilot) Results: DCP Fog removal

* 有 matrix (Deep Learning model) $\rightarrow$ HLS 勝

* 精確訊號處理、有很多時序要處理 $\rightarrow$ Verilog 勝

* Optimization 1: Loop unrolling

* Optimization 2: Array partitioning

* Optimization 3: Resources

* Optimization 4: Pipelining

* Optimization 5: Loop Pipelining

$\rightarrow$ Data Hazard: 對同一個 array 存取

Lightweight Dehazer Circuit System Design for Robot Vision

* Yolo、ViT

$\rightarrow$ 開發影像前處理算法給 AI

$\rightarrow$ 前處理算法在 CPU 跑、Yolo 跑在 GPU

影像前處理算法:

* Low Light Enhancement

* Under Water Image Enhancement

* Dehaze (defog, derain)

如何評估前處理效能?

* 使用 GAN 對 CoCo Dataset 做處理

$\rightarrow$ 再用前處理算法對改造過的 CoCo Dataset 做處理

$\rightarrow$ 處理過的影像 PSNR 只需 20 db 就足以給 Yolo 做 predict